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Physiology, Pulmonary Vasoconstriction

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Introduction

Pulmonary vasoconstriction is a physiological phenomenon and mechanism in response to alveolar hypoxia or low oxygen partial pressures in the pulmonary arterioles and, to some extent, the pulmonary venules. Pulmonary vasoconstriction redirects blood flow within the vasculature away from poorly ventilated parts of the lungs towards better-ventilated portions. Ventilation and perfusion (V/Q) matching is a physiological process that influences gas exchange in the lung, as the lung attempts to efficiently pair oxygenated (ventilated) regions with areas of sufficient blood supply (perfusion). In low-oxygen states, pulmonary vessels constrict in an attempt to shunt blood to better-ventilated regions of the lung. Poor oxygen availability has profound and overarching systemic ramifications manifesting in a plethora of pathologies starting within the lungs itself. Maintaining correct and appropriate oxygen homeostasis is a critical component for systemic stability and functioning, and the process begins within the pulmonary vasculature. While many details pertaining to pulmonary vasoconstriction are not fully understood, the mechanism involves the activity of ion channels as well as several molecular and chemical agents.[1][2]

Issues of Concern

Pulmonary vasoconstriction is a topic of concern as an inappropriate reflex can result in pulmonary hypertension. Chronic pulmonary hypertension may lead to sequelae of hypoxemia, right heart failure, and venous congestion.

Organ Systems Involved

Hypoxic pulmonary vasoconstriction is a pulmonary vasculature mechanism. However, inappropriate and chronic vasoconstriction can lead to pulmonary hypertension, which has systemic ramifications. Pulmonary hypertension increases the work of the heart. The right ventricle, which propels poorly oxygenated blood into the pulmonary artery, works harder to overcome the increasing pressures in the pulmonary circuit, leading to right ventricular hypertrophy in most cases. The grim sequelae of pulmonary hypertension may lead to the classic 'cor pulmonale' presentation in which right heart failure leads to venous congestion.[3][4]

Mechanism

Hypoxic pulmonary vasoconstriction relies on the appropriate functioning and response of the pulmonary vasculature in the presence of diminished oxygen availability. Approximately 250 million alveoli are present within each lung. Each one is a functional unit that serves to deliver inhaled oxygen from the atmosphere to the blood and expulsion of carbon dioxide from the blood to the atmosphere. The alveoli interact with the pulmonary capillaries, allowing for gas exchange.

Theories of the Vasoconstriction Reflex

The vasoconstriction reflex is triggered in states of hypoxia. There are contrasting views as to the vascular structures that first detect hypoxia. The classical understanding suggests that decreased oxygen levels are initially detected within the pulmonary artery, whereas a new concept postulates that low oxygen levels are detected in the alveoli. This latter concept further suggests that gap junctions throughout the pulmonary endothelium transmit signals to the pulmonary arterioles, causing them to constrict.

Original Understanding of the Vasoconstriction Reflex

The original mechanism is thought to involve voltage-gated potassium and calcium channels. These channels are located in the smooth muscle cells of the pulmonary arteries and are very sensitive to low oxygen states. In addition to the critical roles of potassium and calcium in hypoxic pulmonary vasoconstriction, there are indications that there could be other ion channels contributing to the mechanism. These ion channels are transient receptor potential vanilloid 4 (TRPV4) and transient receptor potential canonical 6 (TRPC).[2][5][6]

Related Testing

Echocardiography

Echocardiography uses ultrasound to visualize the four chambers of the heart and provides insight regarding inappropriate internal pressures or hypertrophy. In addition to the size of the right atrium and ventricle, the echocardiogram evaluates right ventricular ejection fraction through the tricuspid annular plane systolic excursion (TAPSE), another important consideration when diagnosing hypertension in the pulmonary vasculature.

Right Heart Catheterization

If preliminary findings warrant further investigation, a cardiologist will conduct a right heart catheterization to confirm the diagnosis. A right heart catheterization provides information regarding the mean pulmonary artery pressure, right atrial pressure, and pulmonary artery occlusion pressure. It also can provide empirical information that would be used to calculate a transpulmonary gradient and pulmonary vascular resistance.

Different Variables and Calculations

In a healthy individual at rest, the mean pulmonary artery pressure (mPAP) is 14 +/- 3 mmHg while a resting (mPAP) of greater than 25 mmHg would meet the criteria for a diagnosis of pulmonary hypertension. Pulmonary artery occlusion pressure (PAOP) and mean pulmonary artery pressure (mPAP) are variables in the calculations for determining transpulmonary gradient and pulmonary vascular resistance.

- The equation to calculate pulmonary vascular resistance: (mPAP-PAOP)/(CO).
- The equation to calculate the transpulmonary gradient: mPAP-PAOP.

Pulmonary artery occlusion pressure is one of the most sensitive findings of right heart catheterization.

Additional Testing

Additional testing needs to be completed to rule out other disease processes that could contribute to pulmonary hypertension. Potential etiologies of pulmonary hypertension include connective tissue diseases, obstructive sleep apnea, embolism, and left heart failure, to name a few. These include spiral CT or ventilation-perfusion (VQ) scans to rule out the presence of thromboembolic activity within the lung's vasculature. Pulmonary function testing identifies and categorizes any intrinsic lung disease that may be contributing. High-resolution CT imaging can be helpful when ruling out pulmonary parenchymal disease.[7][8]

Pathophysiology

Classification of Pulmonary Hypertension

The complete pathophysiology of pulmonary hypertension (PH) is not entirely understood. The current etiologies of pulmonary hypertension are stratified into five groups.

1. The first group is idiopathic PH, likely related to inheritable and intrinsic pulmonary arterial pathology.
2. The second group is PH due to left heart disease, causing a build-up of volume and pressure in the pulmonary vasculature.
3. The third group is PH due to pathology within or related to the lungs. This group includes several different diseases such as COPD, sleep apnea, inflammatory scarring resulting from interstitial lung disease, chronic exposure to a high altitude environment, and certain developmental abnormalities.
4. The fourth group is associated with frequent occurrence of embolic deposition in the lung's vessels or thrombosis known as chronic thromboembolic pulmonary hypertension (CTEPH).
5. The fifth group is designated for any remaining unclear or unestablished etiologies in addition to multifactorial causes. Examples of multifactorial processes that could contribute to or cause PH are metabolic disorders pertaining to the thyroid, hematological disorders, and systemic pathologies that involve the respiratory system, such as sarcoidosis.

Idiopathic Pulmonary Hypertension

Of these groups, idiopathic PH is the least understood. There are two prevailing concepts regarding the progression of idiopathic PH. The most prevalent understanding is related to a derangement in the production of local vasoconstrictors and dilators, denoted as the vasoconstriction theory. The vasodilators include prostacyclin and nitric oxide, and the vasoconstrictors include thromboxane and endothelin.[3]

Roles of Chemical and Molecular Agents in Idiopathic Pulmonary Hypertension

In the vasoconstriction theory, nitric oxide (NO) plays a critical role in maintaining appropriate vascular tone. It is produced and regulated by two enzymes, nitric oxide synthase II and III. Patients with idiopathic pulmonary hypertension are found to have diminished levels of NO. Prostacyclin is a product of arachidonic acid metabolism. It is an effective vasodilator and exerts its effects on platelets as well. In patients experiencing pulmonary remodeling, there are decreased levels of prostacyclin and the enzyme prostacyclin synthase. On the contrary, endothelin 1, a systemic vasoconstrictor, is found in abnormally high quantities in idiopathic PH, which contradicts this theory. [9]

Recent Understanding of Idiopathic Pulmonary Hypertension

A more recent understanding is centered on the concept of endothelial and smooth muscle cell activity in the context of arterial remodeling. Bone morphogenetic receptor type 2 (BMR2) is involved with cell growth, proliferation, and osteogenesis. It is believed that a mutation in BMR2 is directly involved in idiopathic PH. Another contributing factor may involve an inappropriately high sensitivity to depolarization in vascular smooth muscle cells. This alteration in the resting potential setpoint may be due to potassium channel irregularities that cause excess availability of calcium. [10]

Sleep Apnea and Pulmonary Hypertension

A distinct relationship exists between obstructive sleep apnea (OSA) and the presence of pulmonary hypertension (PH). OSA is not accepted as a direct singular cause of PH; however, an association does exist. It is believed that OSA induces hypoxia during sleep, which creates a cycle of oxygen desaturation that leads to decreased nitric oxide synthase, endothelin derangements, and an increase in sympathetic tone.[11]

Clinical Significance

Diagnosis

Hypoxic pulmonary vasoconstriction clinically manifests as pulmonary hypertension. To properly diagnose pulmonary hypertension, clinical criteria must be met in the absence of a primary cause, and the degree of pulmonary hypertension must be determined. Early detection is paramount for long-term morbidity and mortality. The initial process typically begins with an ECG and echocardiogram, followed by pulmonary function tests and cardiac catheterization. Right heart pathology such as hypertrophy should be expected in chronic cases.[12][13]

Nonsurgical Treatment

Therapy is catered to the patient's individual profile. Pharmaceutical regimens are not always indicated for patients; however, anticoagulation is recommended for all diagnoses of idiopathic pulmonary hypertension if there are no contraindications. If pharmacological therapy is indicated, agents such as phosphodiesterase-5 inhibitors, calcium channel blockers, prostacyclin analogs, endothelin receptor antagonists, and cyclic GMP agonists can be utilized. If pulmonary vasoconstriction is refractory to pharmacological therapy or if the severity of the disease is extreme, surgery may be a treatment option. Patients with obstructive sleep apnea should be screened for pulmonary hypertension, and likewise, pulmonary hypertensive patients should be evaluated for sleep apnea due to the association of both diseases. Continuous positive airway pressure (CPAP) may treat OSA and concurrent pulmonary hypertension to a lesser degree.

Surgical Treatment

Surgical options include atrial septostomy, which relieves right heart pressures via a right to left atrial shunt. Lung transplantation may be explored in severe cases.[14][15]

Review Questions

- Access free multiple choice questions on this topic.
- Comment on this article.

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